Supplementary Table Sx. Model fit and evidence of profile-by-age interactions for hospitalisation and chikungunya-attributed death

Model	Age specification	AIC	ΔAIC vs H0	Adult age-interaction test, p	Status
A. Hospitalisation (N = 776,072; 25,363 hospitalisations)					
H0	Common age smooth; no profile-by-age interaction	204,543.2	0 (reference)	—	Reference
H0 (k = 20)	More flexible common age smooth; no interaction	204,510.3	−32.9	—	Sensitivity
H1	Separate age smooth for each profile	204,331.9	−211.3	<0.001	Alternative
H2	**Common age smooth + penalised profile-specific deviations**	**204,302.1**	**−241.1**	**<0.001**	**Selected**
H2 (k = 15)	As H2, larger age basis	204,284.3	−258.9	<0.001	Sensitivity
H3	Common age smooth + age-group × profile random effects	204,365.3	−177.9	—	Alternative
Linear	Common age smooth + profile-specific linear age slopes	204,466.9	−76.3	—	Alternative
B. Chikungunya-attributed death (N = 983,947; 1,050 deaths)					
H0	**Common age smooth; no profile-by-age interaction**	**13,738.6**	**0 (reference)**	**—**	**Selected**
H0 (k = 20)	More flexible common age smooth; no interaction	13,735.4	−3.2	—	Sensitivity
H1	Separate age smooth for each profile	13,723.8	−14.8	0.47	Alternative
H1 (k = 5)	As H1, smaller age basis	13,758.5	+19.9	0.001	Sensitivity
H1 (k = 15)	As H1, larger age basis	13,722.5	−16.1	0.51	Sensitivity
H2	Common age smooth + penalised profile-specific deviations	13,705.9	−32.7	0.67	Alternative
H3	Common age smooth + age-group × profile random effects	13,735.3	−3.3	—	Alternative
Linear	Common age smooth + profile-specific linear age slopes	13,737.6	−1.0	—	Alternative

a. All models are penalised logistic generalised additive models of the outcome on the eight mutually exclusive profiles of diabetes mellitus (DM), hypertension (HTN) and chronic kidney disease (CKD), adjusted for sex, notification year and state of residence (random effect), fitted to the same outcome-specific cohort. Every model includes a non-linear effect of age; models differ only in whether and how profile associations may vary with age. Age bases use k = 10 unless stated.

b. H0, no profile-by-age interaction. H1, a separate penalised age smooth for each profile without a common age smooth. H2, a common age smooth plus penalised profile-specific deviations from it. H3, a common age smooth plus random effects for each age group (0–14, 15–39, 40–59, 60–74, ≥75 years) × profile. Linear, a common age smooth plus a linear age slope for each profile (log-odds scale).

c. ΔAIC = AIC(model) − AIC(H0) within each outcome; negative values indicate better fit after penalising for complexity. AIC values are not comparable between outcomes.

d. Adult age-interaction test: joint Wald test (Bayesian covariance matrix of the penalised fit) of whether the log-odds ratios of the seven profiles relative to None are equal at ages 20, 30, …, 90 years and at the reference age of 60 years. It tests variation in conditional odds ratios, not equality of age-specific risk ratios. — denotes not applicable (H0 models, which contain no interaction) or not directly comparable: H3 has no age-grid test, and for the linear models only a test of the seven slope coefficients is available (p <0.001 for hospitalisation; p = 0.07 for death).

e. AIC and the adult test address different questions: a lower AIC indicates better overall fit, which may arise from childhood or other age ranges, and does not by itself establish that profile associations vary across adult ages.

f. Selected models: H2 for hospitalisation; H0 for death. For death, H1 and H2 had lower AIC than H0, but the adult test provided little evidence that profile associations varied across adult ages (p = 0.47 and 0.67); it was significant only with the restrictive k = 5 basis, which fitted worse than H0 (an inadequate age basis can transfer age misfit into the profile contrasts). Several profiles had few deaths (Supplementary Table Sx+1). H2 (k = 15) had a marginally lower AIC than H2 (k = 10), with similar estimates in adequately supported age groups.

Supplementary Table Sx+1. Observed chikungunya-attributed deaths by multimorbidity profile and age group

Profile	Deaths / cases	Deaths by age group 0–14 / 15–39 / 40–59 / 60–74 / ≥75	Age groups with ≥10 deaths	Age groups with 1–9 deaths	Age groups with 0 deaths
None	*522 / 856,342*	*83 / 113 / 98 / 91 / 137*	*5*	*0*	*0*
DM only	53 / 15,302	0 / 5 / 7 / 9 / 32	1	3	1
HTN only	228 / 79,512	0 / 10 / 34 / 52 / 132	4	0	1
CKD only	12 / 1,640	1 / 1 / 0 / 0 / 10	1	2	2
DM + HTN	181 / 28,379	0 / 5 / 23 / 67 / 86	3	1	1
DM + CKD	4 / 313	0 / 0 / 0 / 0 / 4	0	1	4
HTN + CKD	22 / 924	0 / 1 / 3 / 6 / 12	1	3	1
DM + HTN + CKD	28 / 1,535	0 / 0 / 3 / 8 / 17	1	2	2
Seven non-reference profiles	**528 / 127,605**		**11**	**12**	**12**

Observed chikungunya-attributed deaths in the mortality cohort (N = 983,947; 1,050 deaths) by profile and age group (years). None (reference) is shown for context and is not included in the total row. Of the 35 non-reference profile × age-group strata, 12 had 1–9 deaths and 12 had none. The 10-death threshold is a descriptive indicator of empirical support, not a formal criterion of model validity. CKD, chronic kidney disease; DM, diabetes mellitus; HTN, hypertension.